



Course Name:	Electrical Network Analysis	Course Code:	EE-2004
Program:	BS (EE)	Semester:	Fall 2023
Duration:	180 Minutes	Total Marks:	100
Paper Date:	21-Dec-2023	Weight:	50%
Section:	All	Page(s):	3
Exam Type:	Final	CLO(s)	1,2,3,4,5

Student Name: [REDACTED]

Instruction/Notes:

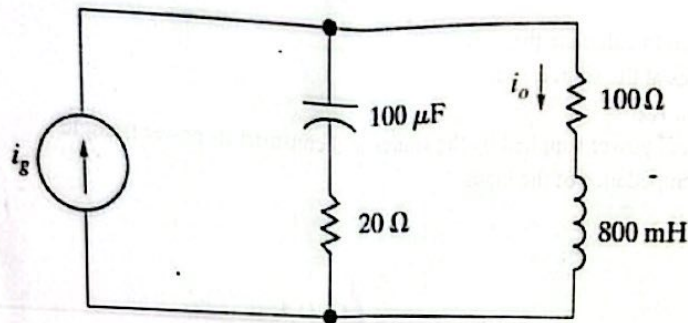
- Attempt all the questions and solve all parts of a question together.
- Carefully understand the question before attempting.
- Show all calculations, direct answers are not acceptable.

Question # 1

CLO 1

[20 marks]

Apply phasor-domain analysis to determine the steady state expressions for i_o if $i_g = 60 \cos(250t)$ mA.



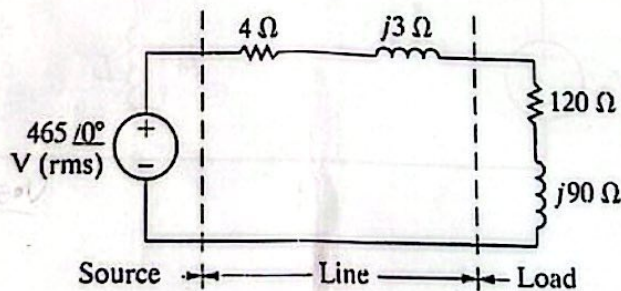
Question # 2

CLO 2

[20marks]

For the circuit shown below, calculate:

- The rms current (I_{rms}) flowing through the circuit.
- The complex power delivered by the source.
- The average power and the reactive power dissipated across the line.
- The average power and the reactive power dissipated across the load. Also, construct its power triangle.
- The value of capacitor, to be included in parallel with the load, to correct load's power factor to unity if the source frequency is 60 Hz.



Question #3

CLO 3

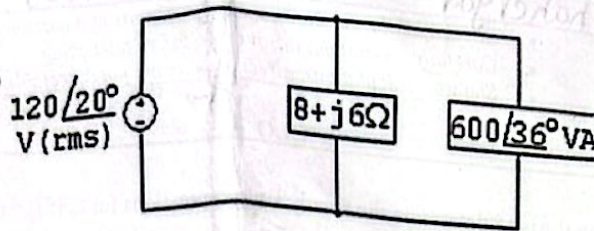
[20 marks]

In a balanced three-phase system, the source is a balanced-Y with an abc phase sequence and a voltage $V_{an} = 120 \angle 20^\circ \text{ V}_{\text{rms}}$. The source feeds two loads, both of which are Y-connected. The impedance of load 1 is $8 + j6 \Omega/\phi$. The complex power for the a-phase of load 2 is $600 \angle 36^\circ \text{ VA}$. The single-phase equivalent circuit for the a-phase is given below.

$(\sqrt{3} \angle 30^\circ) V_{an} = V_{ab}$

$V_{bc} = -120^\circ$

$V_{ca} = +120^\circ$



Analyze the circuit to calculate the:

- line voltages at the source end;
- line current, I_{aA} .
- total complex power supplied by the source and construct its power triangle.
- combined impedance of the loads

Question #4

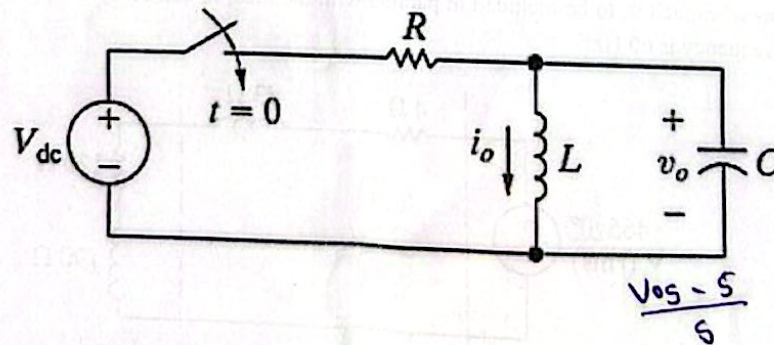
CLO 4

[20 marks]

The switch in the following circuit has been open for a long time. At $t = 0$, the switch closes. Use $R = 2 \Omega$, $L = 0.8 \text{ H}$, $C = 0.25 \text{ F}$, and $V_{dc} = 5.0 \text{ V}$.

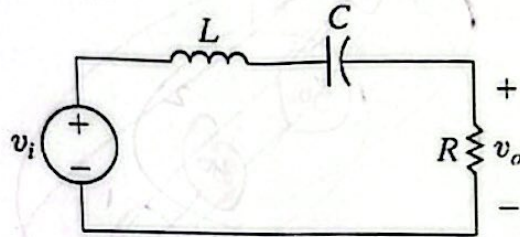
- Transform the given circuit in s-domain, for $t \geq 0$
- Derive the expression of V_o in s-domain.
- Find the time-domain expression of $v_o(t)$.
- Use the s-domain expression derived above to predict the initial- and final-values of $v_o(t)$.

$f = \frac{L}{2\pi} \quad f_s = \frac{2\pi}{2\pi L}$



Consider the following bandpass filter circuit. The filter has a center frequency, f_o of 12.0 kHz and a quality factor, Q of 6.0. Using a $0.1 \mu\text{F}$ capacitor, analyze the circuit to find the:

- Filter's bandwidth, β .
- Values of R and L .
- Filter's upper and lower cut-off frequency, ω_{c1} and ω_{c2} .
- Simplified expression of the filter's transfer function, $H(s) = V_o(s) / V_i(s)$.
- Plot the frequency response of the filter circuit.



$$Z(s) = \frac{1}{sL} + \frac{1}{1/sC} = \frac{1}{0.35} +$$

$$f_c = \frac{1}{2\pi RC}$$

$$f_c = \frac{R}{2\pi L}$$

$$X_L = \omega L$$

$$\omega = 2\pi f$$

$$Z_C = \frac{1}{\omega C}$$

$$\omega =$$